Ethan Krug

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PROFESSIONAL EXPERIENCE

Software Engineer, Wyckoff Farms, Grandview WA Nov. 2023 – Present

- Architected and developed enterprise-level full-stack applications using ASP.NET Core, integrating industry-standard frameworks with robust backend services to deliver scalable solutions serving over 40 field professionals across five departments, resulting in operational efficiency improvements through unified data management and automated workflows.
- Consolidated legacy desktop application and outdated websites into unified receipt processing system for 2 departments, eliminating redundant data entry by approximately 70% and improving financial tracking accuracy.
- Created scalable Financial Hub platform that includes budget management and housing information, featuring quarterly/weekly budget tracking with pivot table visualizations and automated renter balance management.
- Transformed quality control process by creating integrated scoring system, reducing manual data entry and enabling real-time production quality monitoring.
- Centralized organizational resources by creating unified portal for application access and reporting, enhancing operational efficiency across all business units.
- Architected flexible automation platform utilizing WCF services and scheduled DLL execution, enabling seamless integration with external partners and automated data synchronization.
- Implemented scalable image management platform for hops field documentation, featuring optimized storage through compression algorithms and filtering capabilities for cone, bine, and yard images.

Full-Stack Developer (Volunteer), Academics Are Cool, Richland WA June 2021 – July 2023

- Architected modern web platform for statewide mathematics competition management, leveraging React.js and Material UI to deliver intuitive user experiences for students, parents, and administrators
- Engineered scalable data infrastructure using Cloud Firestore, ensuring reliable storage and retrieval of competition registrations, user profiles, and event schedules
- Streamlined competition administration by developing integrated registration system, automating data flow between Google Forms, Sheets, and Firebase backend
- Empowered non-technical staff through intuitive content management interface, reducing dependency on technical expertise for routine website updates and competition announcements

Remote Operations Technician, Strive Tech Inc., Kennewick, WA June – Sept.2019

- Deployed and configured secure banking technology infrastructure across multiple regional branches, ensuring compliance with financial security standards
- Conducted comprehensive site assessments to ensure optimal technology deployment environments, identifying and mitigating potential installation risks
- Provided proactive maintenance and repair services for critical healthcare technology infrastructure, minimizing downtime and ensuring continuous patient care operations
- Delivered targeted training programs for medical and banking professionals, bridging technical knowledge gaps and promoting effective technology adoption

LANGUAGES AND TECHNOLOGIES

- Languages: C#, Java, JavaScript/TypeScript, CSS, HTML, Python, SQL, COBOL, C/C++
- Technologies: ASP.NET Core, MS SQL Server, Git, GitHub, Visual Studio, VS Code, React.js, Material UI, jQuery, Google Cloud (GCP), WCF Services
- Other: Code Optimization, API Design, System Architecture, Performance Engineering, Testing Frameworks, CI/CD Pipelines, Agile Methodologies

EDUCATION

Bachelor of Science in Computer Science
Minor: **Mathematics**
Central Washington University, Ellensburg, WA
Graduated Magna Cum Laude June, 2022

PROJECTS

Personal Portfolio Website April 2023 – Present

- Architected dual implementations of personal portfolio website using React/Material UI with TypeScript and vanilla JavaScript/CSS, demonstrating versatility in frontend development approaches
- Engineered custom Content Management System using GitHub API for blog post management and content versioning
- Developed Python-based data generation system for JSON creation and blog content integration
- Implemented RSS feed functionality with XML generation, expanding site discoverability and content

Algorithm Visualization Platform Nov. 2022 – Sept. 2023

- Designed and developed interactive visualization tool for path-finding and graph-search algorithms using Python
- Architected and implemented comprehensive algorithm suite with comparative analysis capabilities
- Created automated testing framework for algorithm performance benchmarking and validation
- Led cross-platform migration effort, translating desktop application to web-based React/TypeScript implementation

KiwiLink Mobile Application, Seattle, WA Dec. 2019 – July 2020

- Collaborated with distributed team of 12+ developers on enterprise mobile application
- Spearheaded UI/UX design initiatives using Figma, driving consistent design language adoption
- Developed core profile functionality using React Native, implementing responsive layouts and component architecture

CWU Web Application Project, Ellensburg, WA Jan. – March 2022

- Led Agile development team in collaborative web application development
- Architected user interface components using React.js and Material UI framework
- Facilitated daily SCRUM ceremonies and weekly stakeholder meetings, ensuring transparent project progression
- Coordinated with project manager to identify and resolve technical blockers, maintaining project timeline integrity